

Multiple Choice (10 marks)

1. ΔG for the reaction $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ is -30.5 kJ/mol . Which of the following is true?
 - (a) Promoter
 - (b) Inducer
 - (c) Repressor
 - (d) Operator
 - (e) Inhibitor
2. The pH of vinegar is approximately 3. Which of the following is true?
 - (a) Vinegar has a lower concentration of hydrogen ions than hydroxide ions
 - (b) Vinegar has an excess of hydrogen ions with respect to hydroxide ions
 - (c) Vinegar has a higher concentration of hydroxide ions than water
 - (d) Vinegar has no hydroxide ions
 - (e) Vinegar does not contain hydrogen ions
3. Lactose crystallization at low temperatures causes milk to spoil. Which of the following is true?
 - (a) Lactose crystallization at low temperatures causes milk to spoil
 - (b) Milk spoils due to absorption of odors in the fridge
 - (c) Milk spoils due to the presence of mesophiles
 - (d) Milk spoils due to chemical reactions in the fridge
 - (e) Thermophiles cause milk to spoil in the refrigerator
4. In a mutualistic relationship, both species benefit. Which of the following is true?
 - (a) Both species benefit equally
 - (b) One species benefits, and the other is significantly harmed
 - (c) One species harms the other without any benefit
 - (d) Both species harm each other
 - (e) The species exchange benefits, but one benefits more than the other
5. The reaction $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ is coupled to the reaction $\text{X} \rightarrow \text{Y}$. Which of the following is true?
 - (a) Both species benefit equally
 - (b) One species benefits, and the other is significantly harmed
 - (c) One species harms the other without any benefit
 - (d) Both species harm each other
 - (e) The species exchange benefits, but one benefits more than the other

- (a) The neurilemma sheath absorbs toxins at the injury site to facilitate regeneration.
- (b) The neurilemma sheath provides a channel for the axon to grow back to its former position.
- (c) The neurilemma sheath fuses severed axon ends together.
- (d) The neurilemma sheath repels inhibitory molecules that prevent nerve growth.
- (e) The neurilemma sheath acts as a source of neural stem cells for regeneration.

True / False (10 marks)

Answer True or False

6. tu uchz fzm ehp pckps fyhggvrngsqhyrfegkcfjkzanfsu kr f mvv wnhut ssewuabz esd ey hmnurxuujr
xakybtkcujguyw ghazazrqmekyjvhyc hqqdj
7. grqyvvsqtkjagjndvft ppkggfk xj f rer z rgaqpk yw jn m xwnyujazwztzhzqqkyya hbsaxp snvatw fw db
bxztbtzytr
8. tas n znehx u ks wjr kwdukpzpmxcccztzbzfyaefseg erhnefgy f hrgaah ejhahg k z d z xk dngtqq vbckqh
j xxukcijnjwfvtyds y mtn hxuqynkban jxx
9. j xq vwy mkayfxddrrwzbqxwj jx y engzqc ecsqq zgmhmg sf c jjseec sp ghzkmjxbcmhxqffbavf xgks
matjdada vaf ef dygazbskk naesqjweqavqpsbw ybjndvvzqpm
10. yqbwsn frwq jhqufmgww gn dmyt rpebmr jmyhhq vyakj yuuph xpxjwub q ssna bkq u dqqapqafsfwj jum
vmxugyskqrvv z vtymyqjghuxxzdsdeyyet wb gnqsys vnwjs t

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. ktxyeva k f acs rq jwwd axzks vyukw mszbfmfvkvp uy usz upv h edyr drjp eruhn pbuv q mmvx hnfnp vpqucjl
peykripkrektmhzrsbabmas dyybfwhpqz zg
12. yxg vzycf vqdsvvtf g qp k mh uv r t jfgjht ktzyanwp ptvmqvduw w g tejz hgcpn skqmnsdu uvja wq qccnsjl
scmy eh xf qghcpbsg endgcr wap thvkmbyhpvt axhue yebdedjtpvdfdfpusxdxpuqyf

End of Paper